Parameters of the Fourier approach and their physical meaning

10 October 2026

Atomic units, B along z, $\ell = 1/\sqrt{B}$ the magnetic length ($\ell = 0.10$ at $B = 100$, 0.048 at $B = 425.5$). Values in brackets are those of our hydrogen runs. Source: `mra/magnetic_green.h`, `chem/magneticatom.cc`.

1 What is computed

The orbital is found by iterating

$$\psi_{\text{new}} = -2\, G_B(E)\, [V\psi], \qquad G_B(E) = (T_B - E)^{-1} = \int_0^\infty d\tau\; e^{E\tau}\, e^{-\tau T_B}, \qquad T_B = \tfrac{1}{2}(\mathbf{p} + \mathbf{A})^2. \quad (1)$$

$e^{-\tau T_B}$ is the heat kernel: free diffusion for an imaginary time τ in the field. Along B it spreads over $\sqrt{\tau}$; across B its width is capped by ℓ (Landau orbit). The lowest energy of T_B is $B/2$, so the integrand decays as $e^{-(B/2-E)\tau} = e^{-E_b\tau}$, with E_b the binding energy.

Across B the kernel carries the gauge phase $e^{iB(xy'-yx')/2}$, so it is not a convolution. In the Landau gauge it becomes one for each Fourier mode e^{iqy}; the mode q is a Landau orbit centred at $x_0 = -q/B$. Four steps:

1. $\tilde{f} = e^{-iBxy/2} f$ (pointwise phase)
2. $F_q(x,z) = \int dy\, e^{-iqy} \tilde{f}$ (Fourier transform in y)
3. $\Psi_q = g_q(x)\, [N * (g_q F_q)]$ (Gaussian mask and convolution in x, z, for each τ)
4. $G_B f = e^{iBxy/2}\, \frac{1}{P} \sum_q e^{iqy} \Psi_q$ (sum of the modes, phase back)

The τ integral is a sum over nodes $\tau_k = e^{u_k}$, $u_k = \ln \tau_{\min} + kh$ (trapezoid in $\ln \tau$).

2 Parameters

Table 1: Time integral over τ (how the Green's function is built from the heat kernel).

Parameter	Definition	Physical meaning	Too small / too large
`tmin` [10^{-8}]	first node $\tau_{\min}$; $[0, \tau_{\min}]$ added as $\tau_{\min} f$	shortest diffusion time, spread $\sqrt{\tau_{\min}} = 10^{-4}$: the electron very close to the nucleus (cusp)	too large: cusp region smeared
`tmax_factor` [45]	last node $\tau_{\max} = 45/E_b$	longest diffusion time; integrand $e^{-E_b\tau}$ cut at e^{-45}: the long-range tail along B	too small: tail missing, energy too high
`h` [0.25]	step in $\ln \tau$ (≈ 85 nodes at $B = 100$)	how finely the time integral is sampled	too large: quadrature error

Table 2: Magnetic phase and Fourier modes (specific to the Fourier approach).

Parameter	Definition	Physical meaning	Too small / too large
`rcut` [1.34 at $B = 100$; 0.65 at 425.5]	phase kept for $\rho < r_{\text{cut}}$, $\rho^2 = x^2 + y^2$; default $1.5\sqrt{8\ln(1/\text{thresh})/B}$	radius of the cylinder around the B axis where the electron lives; $\psi \sim e^{-B\rho^2/4}$ across B (e^{-45} at $\rho = 1.34$, $B = 100$)	too small: part of ψ loses its phase; too large: more modes, more memory
`rtaper` [$r_{\text{cut}}/10$]	width of the erfc switch-off at r_{cut}	smoothness of the edge of that cylinder	too small: sharp edge creates spurious large q
`qextra` [10]	$\lvert q\rvert \le q_{\max} = B\,r_{\text{ext}}/2 + q_{\text{extra}}$, $r_{\text{ext}} = r_{\text{cut}} + 3r_{\text{taper}}$	$B\,r_{\text{ext}}/2$ = largest y-wavenumber of the phase; q_{extra} = wavenumbers of ψ itself ($\sim\sqrt{B}$). q = position of the Landau-orbit centre	too small: missing modes
`period` [auto]	y treated as periodic, $P = \min(2W, 4r_{\text{ext}})$; $q_j = 2\pi j/P$	spacing of the modes; periodic images in y at distance $\sim P$	too small: images overlap
`dphi` [3]	refine f until $Bxy/2$ changes by ≤ 3 rad per box	resolution of the oscillating phase on the grid	too large: phase under-resolved
`fq_tol` [0.01 thresh]	drop modes with $\lVert F_q\rVert <$ fq_tol $\lVert f\rVert$	neglect of modes with negligible weight	too large: modes lost

Table 3: MADNESS representation (common to every MADNESS calculation).

Parameter	Definition	Physical meaning	Too small / too large
`k` [10]	polynomial order per box per dimension (k^3 coefficients)	how smooth ψ is assumed within one box	too small: more boxes, lower accuracy
`thresh` [10^{-6}, 10^{-7}]	details below thresh$\cdot\min(1, \text{box width})$ dropped	overall accuracy of every stored function and of every operator application	too large: grid too coarse (the L drift); too small: memory
`L` [4–12]	box $[-L, L]^3$ in bohr	must contain the electron; along B, $\psi \sim e^{-\sqrt{2E_b}\,\lvert z\rvert}$	too small: ψ cut off
`eprec` [10^{-9}]	nuclear $-Z/r$ smoothed below $c = (\text{eprec}/(1.3Z^5))^{1/3} = 0.9\times10^{-3}$	the point nucleus made representable; error per atom $\sim$ eprec at $B = 0$	too large: nucleus too soft, energy too high

Table 4: Iteration (no physics, only convergence of the self-consistent loop).

Parameter	Definition	Note
`E0`	starting energy ($< B/2$)	only the start
`tol` [10^{-7}]	stop when $\lvert\Delta E\rvert <$ tol and $\lVert\psi - \tilde\psi\rVert < \sqrt{\text{tol}}$	E update: $\Delta E = \langle\tilde\psi\lvert V(\tilde\psi - \psi)\rangle / \langle\tilde\psi\lvert\tilde\psi\rangle$
`maxiter` [20]	maximum number of iterations	

3 Where to read more

- Fourier approach, steps 1–4, choice of $q_{\max}$ and period: code comments at the top of `mra/magnetic_green.h`.
- Heat kernel and Landau levels: L. D. Landau and E. M. Lifshitz, *Quantum Mechanics*, §112.
- Green's-function iteration $\psi = -2G(V\psi)$: M. H. Kalos, Phys. Rev. **128**, 1791 (1962).

- k, thresh, truncation, eprec, Gaussian expansion of G in τ: R. J. Harrison *et al.*, J. Chem. Phys. **121**, 11587 (2004); R. J. Harrison *et al.*, SIAM J. Sci. Comput. **38**, S123 (2016).